



# Mohamed Yahya SELLAMI

Junior Full-Stack & IoT Developer | Software Testing  
Bachelor's Degree in Computer Science and Information Systems

## PROFILE

Computer Science and Information Systems graduate with hands-on experience in full-stack development, IoT systems, and software testing. Experienced in API, integration, and system testing across backend, web, mobile, and embedded environments. Skilled in Spring Boot, PostgreSQL, React, Flutter, C/C++, REST APIs, MQTT, and hardware-software integration. Strong understanding of end-to-end system behavior with a focus on reliability, software quality, and problem solving.

## ACADEMIC TRAINING

Engineering Cycle in Computer Science – In Progress  
Institut International de Technologie – IIT (Evening classes)



2026 – Present

Bachelor's Degree in Computer Science and Information Systems  
Institut International de Technologie – IIT



2023 – 2026

Baccalaureate in Technical Sciences – Lycée Ibn Rachik, Sfax

2023

## PROFESSIONAL INTERNSHIP

BlueBox Labs — Full-Stack & IoT Developer Intern



02/2026–06/2026

Contributed to the end-to-end development of the BlueBox smart irrigation platform, including backend services, web and mobile applications, and embedded controller integration.

Main responsibilities:

- Developed REST APIs and backend services using Spring Boot and PostgreSQL for device management, irrigation programs, logs, and real-time status updates.
- Built React web dashboard features for controller monitoring, device configuration, reporting, and irrigation management.
- Developed Flutter mobile application features for controller onboarding, remote valve control, irrigation scheduling, and real-time monitoring.
- Integrated MQTT and REST/JSON communication to synchronize data between IoT controllers, the backend, and user applications.
- Integrated Firebase Cloud Messaging for system alerts and user notifications.
- Developed and improved ESP32-S3 firmware features using C/C++.
- Implemented controller logic for irrigation control, state synchronization, sensor monitoring, and real-time status reporting.
- Performed API, integration, and system testing across backend, web, mobile, and IoT components.
- Tested REST APIs and JSON data exchanges between Flutter/React applications, Spring Boot backend services, and ESP32 controllers.
- Identified and investigated functional and integration issues to improve overall platform reliability.

### SECOMÉ:

Learned and applied the React framework and Spring Boot to design and develop a full-stack web application, integrating a responsive front-end with a secure and efficient back-end, improving overall user experience and system performance.



06/2025–08/2025  
Sfax, Tunisia

### FUTURE PROOF:

Learned and applied the React framework to develop and enhance dynamic web interfaces during the internship.



08/2024–09/2024  
Sfax, Tunisia

### I TECH:

Gained practical knowledge of computer hardware components and their installation, configuration, and troubleshooting.



06/2024–07/2024  
Sfax, Tunisia

## PROJECT

### SMART AUTONOMOUS VEHICLE:

Collaborated with a team in the Smart Autonomous Vehicle Challenge to design and develop an autonomous vehicle using a Raspberry Pi board. Contributed to programming the system for navigation and obstacle detection through sensors, enhancing my skills in embedded systems, hardware, and teamwork.

10/2025–11/2025  
Sfax, Tunisia

### AFTER-SALE SERVICES WEB APPLICATION:

Developed a web application using .NET and SQL Server for managing After-Sale Services.

5/2025–6/2025  
Sfax, Tunisia

Built separate interfaces for administrators and clients with secure data management.

Focused on scalability, clean architecture, and an intuitive user experience.

## ASSOCIATIVE EXPERIENCE

IEEE, Mentor of IEEE RAS IIT SBC

1/2026–now

IEEE, Chair of IEEE RAS IIT SBC

1/2025–12/2025

scout, Scout leader

2019–now

IEEE, project manager of IEEE RAS IIT SBC

5/2024–12/2024

## contact



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## languages

Arabic (native)

English (B1)

French (B1)

## technical skills

### Software Testing & QA:

API Testing, Functional Testing, Integration Testing, System Testing, Regression Testing

### Programming Languages:

C, C++, Python, Java

### Embedded & IoT:

Embedded Systems, Firmware

Development, MQTT, REST APIs, Wi-Fi, JSON

### Web & Mobile Development:

Flutter, HTML, CSS, JavaScript, React

### Backend Development:

Spring Boot, .NET

### Databases:

MySQL, Oracle, SQL Server, PostgreSQL

### Tools:

Git, GitHub/Bitbucket, Docker, PlatformIO, Android Studio

## soft skills

Leadership and team motivation

Strong decision-making and problem-solving skills

Strong stress management skills

Clear and effective communication within a team